

MARBLEHEAD, OH, USA

WMO: 997293

Lat: 41.544N

Lon: 82.731W

Elev: 623

StdP: 14.37

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.6	(c) 7.2	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 30.2	(k) 22.7	(l) 27.6	(m) 23.2	(n) 8.4	(o) 220	(p) 0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.6	(c) 88.5	(d) N/A	(e) 85.2	(f) N/A	(g) 82.3	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 6.6	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 27.8	(b) 24.3	(c) 20.6	(e) -1.7	(f) 93.5	(g) 7.5	(h) 2.4	(i) -7.1	(j) 95.2	(k) -11.5	(l) 96.6	(m) -15.7	(n) 98.0	(o) -21.1	(p) 99.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.5	26.0	26.8	36.4	48.6	60.4	70.1	74.8	73.2	67.2	55.4	42.9	34.5
	DBStd	19.03	11.15	10.78	9.57	8.54	8.14	5.93	4.93	4.21	6.34	7.75	8.71	8.93
	HDD50	2728	746	651	437	125	7	0	0	0	33	243	486	
	HDD65	5846	1209	1069	886	496	196	19	1	2	46	314	662	946
	CDD50	3273	2	1	16	84	328	604	768	719	516	199	31	5
	CDD65	918	0	0	0	5	52	174	304	255	112	16	0	0
	CDH74	4958	0	0	0	56	294	902	1993	1192	476	45	0	0
	CDH80	1154	0	0	0	7	74	227	507	207	125	7	0	0
(14) Wind	WSAvg	8.6	9.5	9.1	10.1	10.4	8.5	7.8	7.0	7.4	8.0	8.7	8.2	9.0
(15) Precipitation	PrecAvg	34.1	1.8	1.7	2.6	2.9	3.5	3.9	3.7	3.5	3.1	2.1	2.8	2.6
	PrecMax	45.0	4.6	4.2	5.4	7.2	5.2	10.3	12.6	9.9	6.7	4.6	6.5	6.5
	PrecMin	18.3	0.2	0.0	0.9	0.9	0.7	0.7	1.0	0.7	0.9	0.0	0.2	0.8
	PrecStd	5.7	1.2	1.0	1.2	1.4	1.1	2.0	2.3	1.9	1.6	1.1	1.6	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.7	58.4	68.6	80.9	86.8	90.0	92.2	89.6	89.5	78.6	67.3	61.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	53.3	51.6	62.4	73.8	81.3	86.1	88.8	84.8	84.0	74.3	62.6	55.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	47.0	45.3	57.2	68.6	76.9	82.0	85.5	81.9	78.9	70.5	59.5	51.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	40.9	39.6	50.9	62.6	72.7	79.1	82.4	79.7	76.0	66.4	56.1	46.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	11.5	11.5	11.2	12.9	12.3	10.9	10.6	10.0	10.5	11.1	10.5	9.9
		MCDBR	16.9	19.2	20.2	23.9	19.4	16.4	14.9	13.8	15.7	15.5	16.3	15.9
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.309	0.333	0.358	0.398	0.434	0.447	0.442	0.432	0.387	0.365	0.326	0.315
	taud		2.395	2.307	2.313	2.256	2.199	2.211	2.245	2.261	2.364	2.403	2.474	2.427
	Ebn at Noon		266	275	281	277	268	264	264	263	269	261	258	253
	Edh at Noon		26	33	36	42	45	45	43	41	35	30	24	23
(44) All-Sky Solar Radiation	RadAvg		451	714	1084	1456	1721	1916	1929	1693	1365	889	561	378
	RadStd		46	98	95	113	144	128	108	64	117	69	82	40

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+180	N/A

Nomenclature: See separate page